Buck Converter Design

Designing and implementing a Buck Converter in MATLAB Simulink.

The overall goal here is to implement a Buck Converter from scratch. Design goal is to reliably step-down 12V to 5V, a typical application (car USB chargers, powering microcontrollers).

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1. Specifications

$V_{in} = 12V$, range 10-14 V
$V_{out} = 5V$
$W_{out} = 20W$
$I_{load} = 4A$
$f_{sw} = 100kHz$
$\Delta I_L = 1A$ (25% of I_{load})
$\Delta V_{out} = 50mV$ (1% of V_{out})

2. Calculation of L, C, duty cycle range and CCM boundary

Using worst case $V_{in} = 14V$ for sizing of the inductor to achieve desired current ripple for full range of input voltage.

$$L = \frac{V_{out}}{\Delta I_L \, f_{sw}} \left(1 - \frac{V_{out}}{V_{in}}\right)$$

$L = 3.214 \times 10^{-5} H$

Picking standard value of $33\mu H$ is a safe choice.

Checking the ripple with $V_{in} = 10V$:

$$\Delta I_L = \frac{V_{out}}{L\, f_{sw}} \left(1 - \frac{V_{out}}{V_{in}}\right)$$

$\Delta I_L = 0.758A$

Duty cycle D for the range of V_{in} :

$$D = \frac{t_{on}}{T} = \frac{V_{out}}{V_{in}}$$

$D = 0.5$ for $V_{in} = 10V$
$D = 0.417$ for $V_{in} = 12V$
$D = 0.357$ for $V_{in} = 14V$

Capacitor sizing:

$$C = \frac{\Delta I_L}{8\, f_{sw}\, \Delta V_{out}}$$

$C = 2.5 \times 10^{-5} C = 25\mu F$

Considering ESR effects, **reasonable pick is $C = 47\mu F$ with $ESR = 10m\Omega$**,
giving roughly 27 mV capacitive and 10 mV ESR ripple, leaving space below 50 mV.

$$\Delta V_{ESR} = \Delta I_L \cdot ESR$$

CCM Operation:

CCM boundary is reached when $I_{Lmin} = 0$

$$I_{Lmin} = I_{load} - \Delta I_L/2$$

$$I_{load,min} = \frac{\Delta I_L}{2}$$

The worst case is at Vin = 14V and $I_{load} = 0.5A$.

Thus, the converter enters DCM below 2.5W.

3. Open-loop power stage design in MATLAB Simulink

PWM Generator is built using a Sawtooth Generator (100kHz frequency) with a comparator and a constant duty value of D = 0.417 (For 12V input voltage).

For switching, I use a MOSFET with threshold voltage of 2V and drain-source on resistance of 0.01Ω
Gain of 3 is required to trigger MOSFET's threshold.

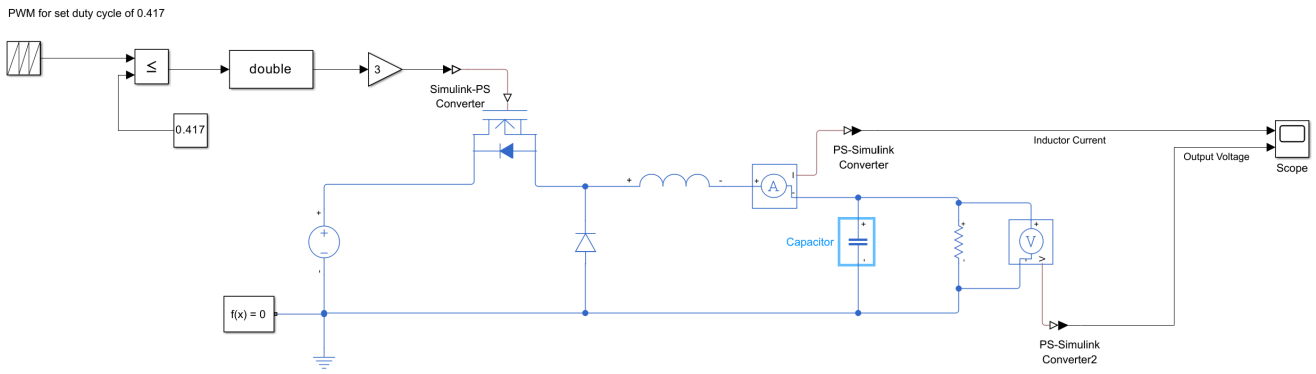


Figure 1. Open-loop power stage Simulink model.

Using standard diode from the Simscape Electrical library with forward voltage of 0.6V and on-resistance of 0.3Ω gives a large loss on that diode, bringing output voltage to about 4.07V (in simulation) instead of desirable 5V.

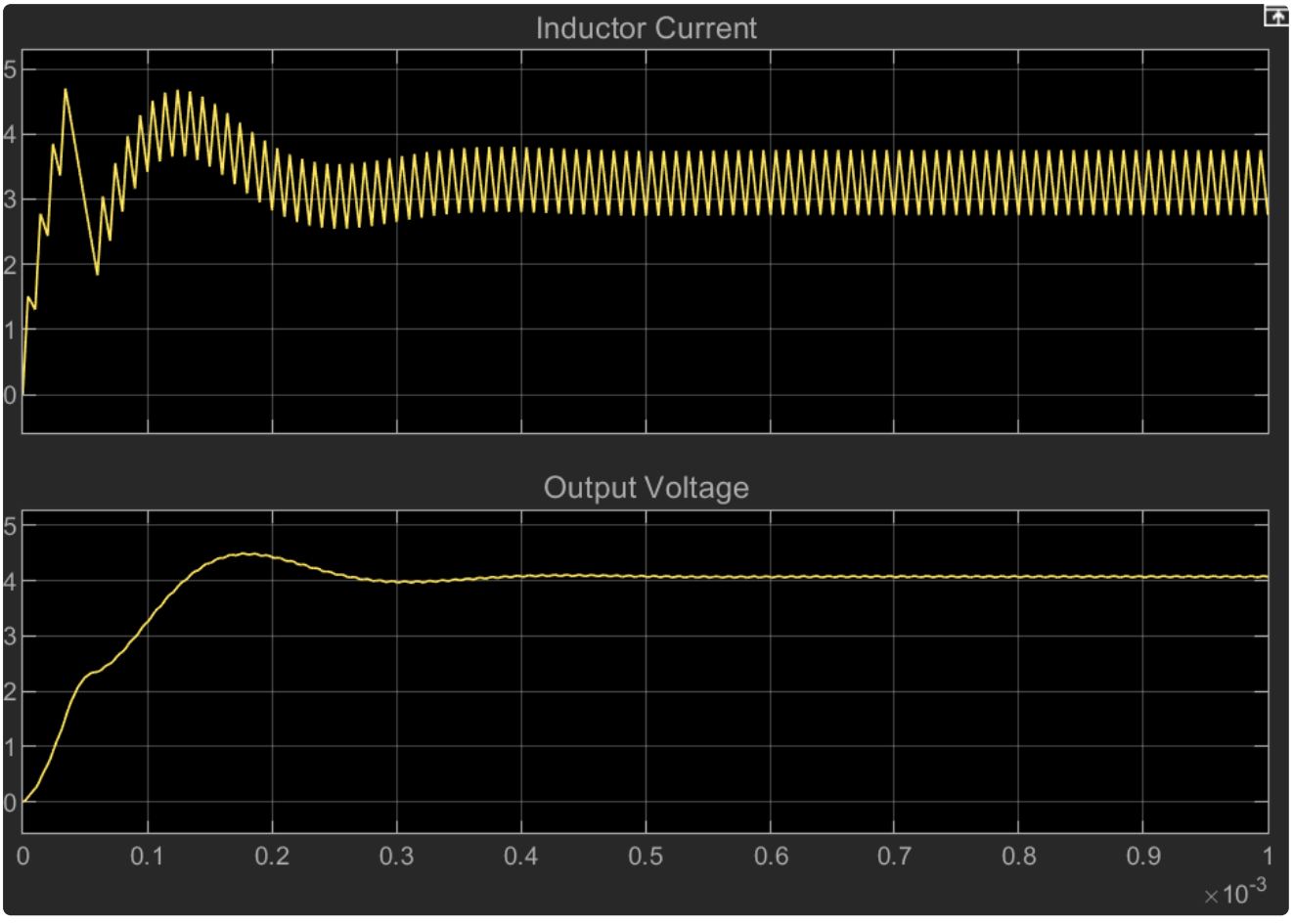


Figure 2. Inductor Current (A) and Output Voltage (V) readings for the standard diode model in open-loop stage.

It is useful to account for that diode in calculation to check whether the simulation is accurate.

Volt-second balance equation:

$$(V_{in} - V_{out})D + (-V_{out} - V_f - R_{on}I_{load})(1 - D) = 0$$

$$V_{out} = DV_{in} - (1 - D)(V_f + R_{on}I_{load})$$

Measured I_{load} is approximately 3.26A.
Solving for V_{out} yields:

$V_{out} = 4.08V$

Which almost perfectly matches with the simulation.

Now let's be more realistic and use a Schottky Diode with a forward voltage of 0.35V and on resistance of 30 $m\Omega$ (e.g. MBRS540T3G).

Calculated prediction is (substituting I_{load} with $\frac{V_{out}}{R}$):

$V_{out} = 4.73V$
$I_{load} = 3.79A$

Comparing with the simulation:

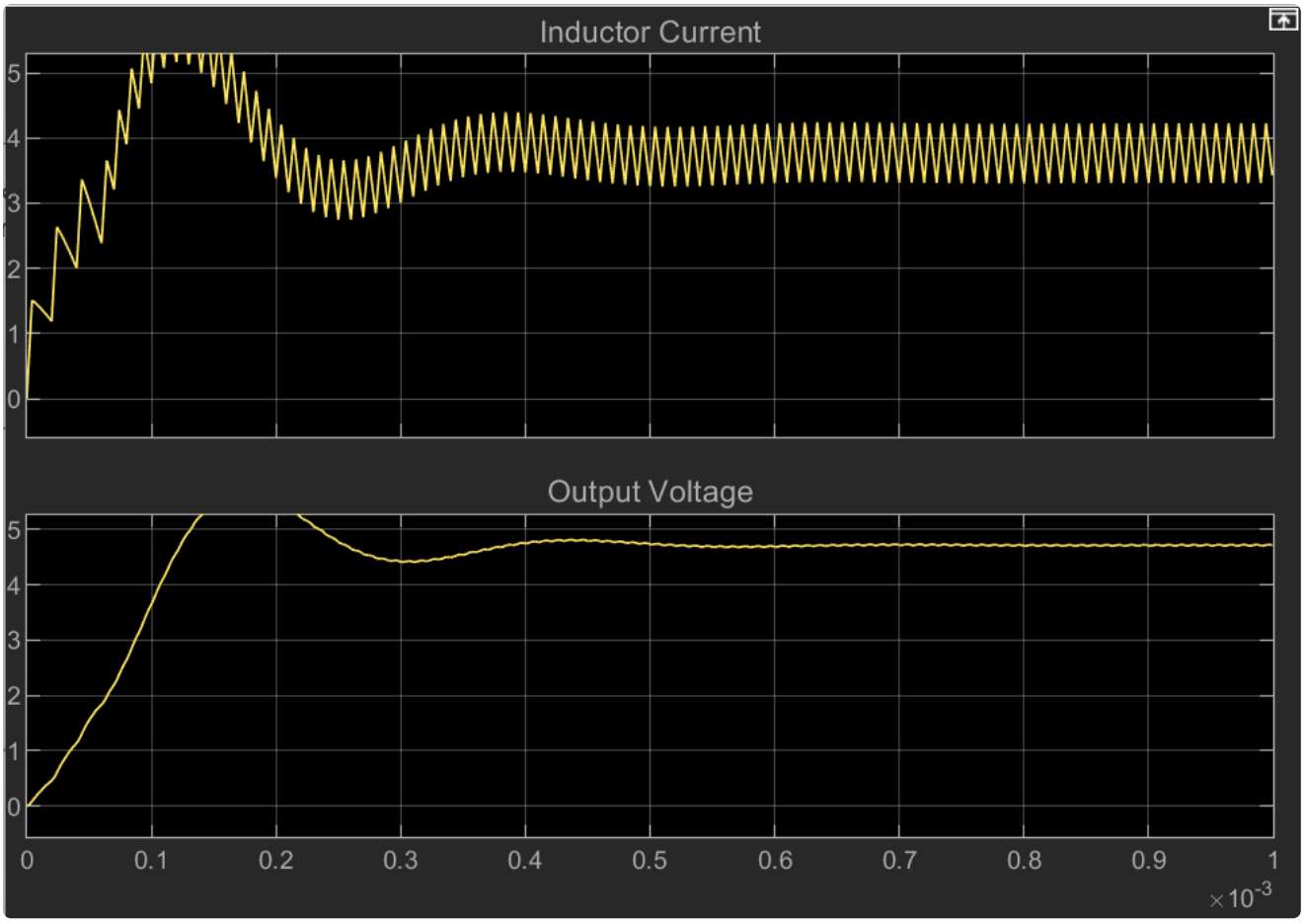


Figure 3. Inductor Current (A) and Output Voltage (V) readings for the chosen Schottky diode model in open-loop stage.

Results match almost perfectly, and it is a good improvement from earlier 4.07V.

To get even better results, a controller is needed.

4. Control Design

I choose the voltage-mode control that adjusts the duty cycle to control output voltage.
It is simple yet effective, and a good starting point.

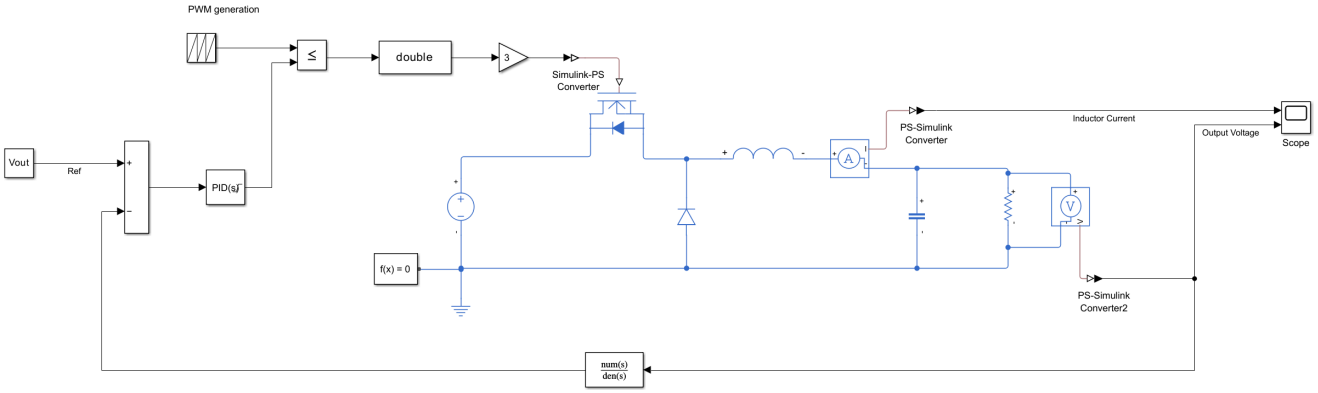


Figure 4. Closed-loop control design of a buck converter, Simulink model.

Controller is designed based on a small-signal averaged model, assuming CCM, ideal switch and ideal diode.
Here, I neglect the diode forward drop because it mainly does DC offset which is managed by integral action of the PID.

Plant transfer function:

$$G(s) = V_{in}\frac{1 + sC * ESR}{1 + s(L/R + C * ESR) + s^2 LC\frac{R+ESR}{R}}$$

Measured output voltage is passed through a low-pass filter at 20 kHz to attenuate the switching ripple.

$$H(s) = \frac{1}{1 + \frac{s}{2\pi \cdot 20\ \text{kHz}}}$$

Controller design is aimed for 5kHz crossover frequency and 60 degrees phase margin (at nominal 12V input and 4A load), and a requirement to operate stable across the input voltage range: 10 to 14V.

4.1 Tuning

Which was the hardest part, as MATLAB's pidtune managed very well at one point, but when moving to another point it quickly became a disaster. This is explained by load-dependent damping of the LC filter, which fails under light load.

The final controller was tuned by manual placement following Type III placement approach:

- Double zero at half the resonance frequency, 2.02 kHz, which provides a phase boost through the resonance region, independent of how sharp the resonance is, instead of

trying to cancel it.

- Pole at 50 kHz, limits the high-frequency gain, ten times above crossover.
- Gain K = 382, so the loop crosses over at exactly 5 kHz at the nominal point.

4.2 Robustness Check

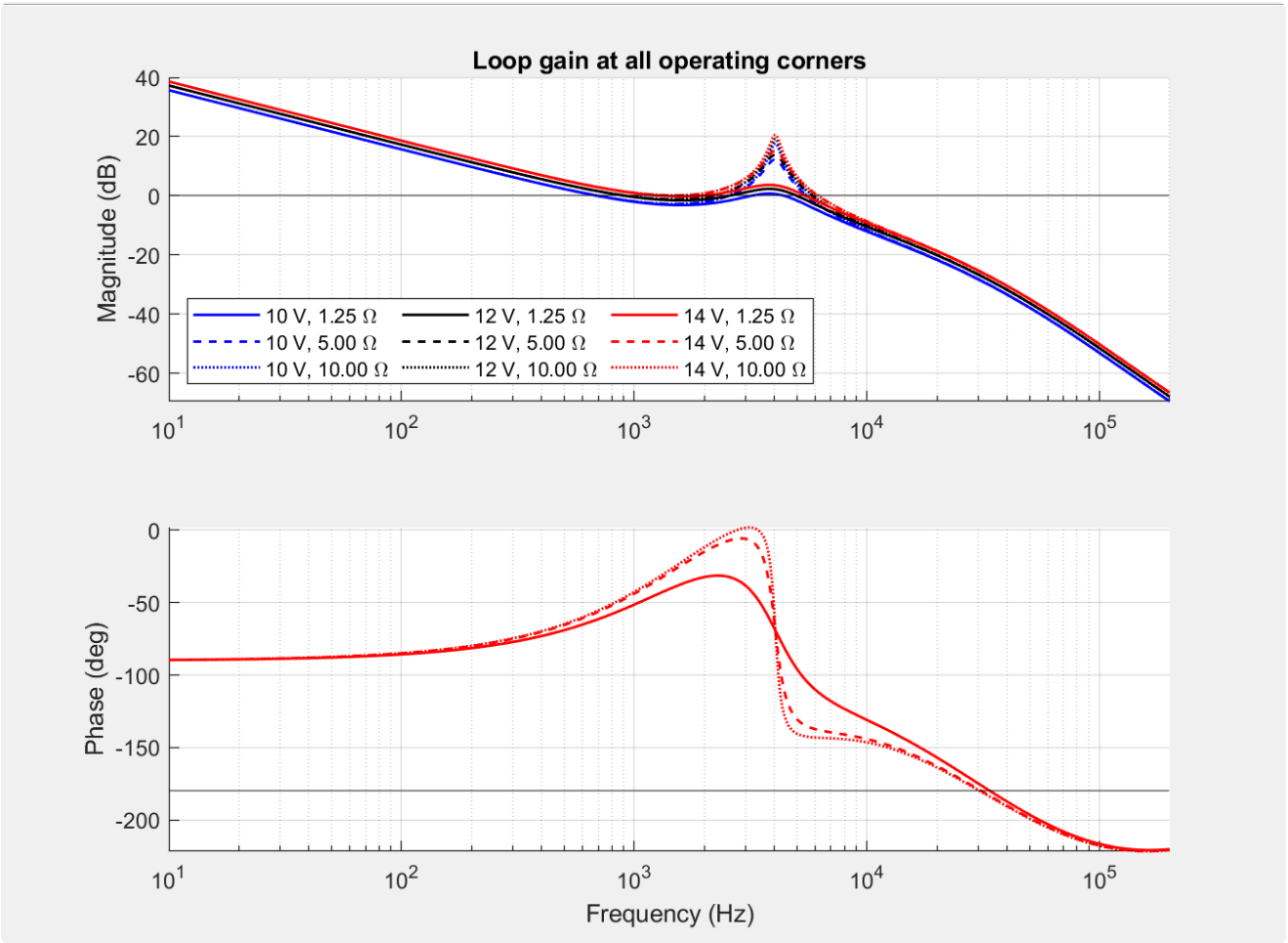


Figure 5. Loop gain at nine operating corners (10–14 V, 1.25–10 Ω).

Table 1. Corner checks

$V_{in}[V]$	$R[\Omega]$	$PM[deg]$	$f_c[kHz]$	$GM[dB]$
10	1.25	98.7	4.42	30.3
10	5.00	44.8	5.50	28.7
10	10.00	37.5	5.55	28.4
12	1.25	84.3	5.00	28.7
12	5.00	43.3	5.83	27.1
12	10.00	37.1	5.87	26.8
14	1.25	76.7	5.45	27.4
14	5.00	42.2	6.15	25.8
14	10.00	36.8	6.19	25.5

The crossover stays between 4.4 and 6.2 kHz everywhere, the phase margin never falls below 36.8°, and the gain margin stays above 25 dB.

5. Verification in the Simulation

Tuned controller was verified on the Simulink Simscape Electrical model (Figure 4).

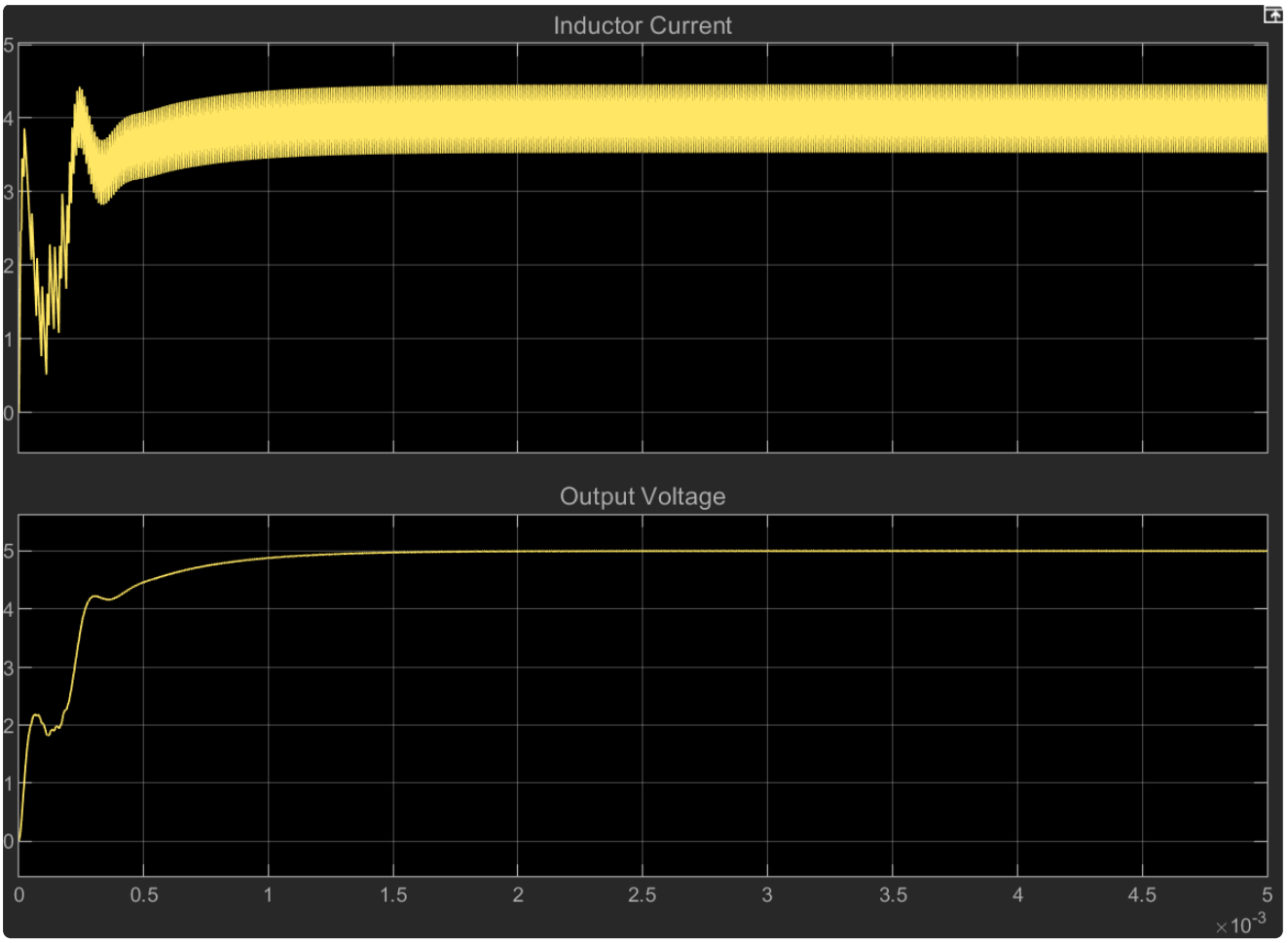


Figure 6. Inductor Current (A) and Output Voltage (V) in the closed-loop control system with nominal input voltage and load.

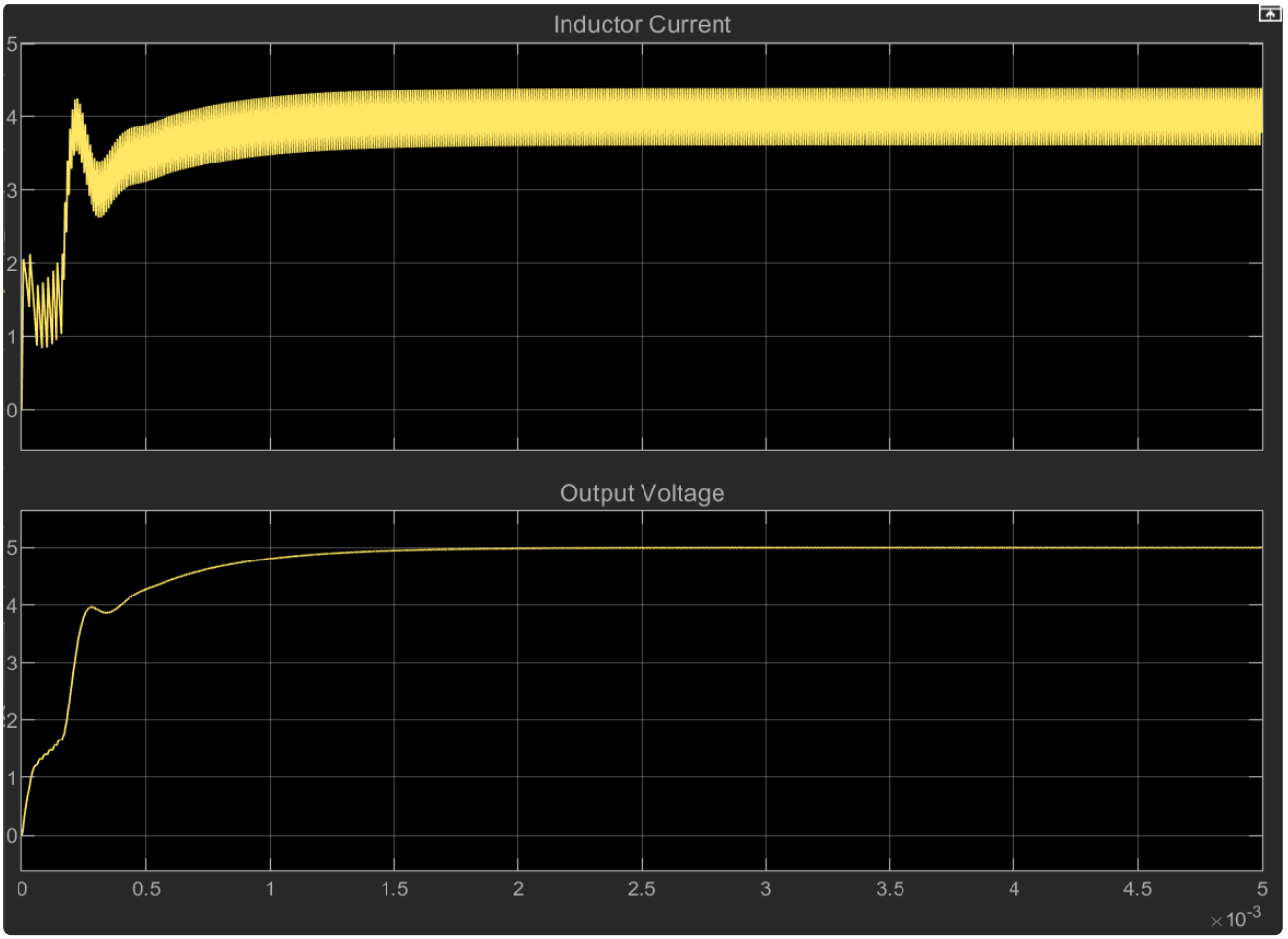


Figure 7. Inductor Current (A) and Output Voltage (V) in the closed-loop control system with input voltage of 10V (low side of range) and nominal load.

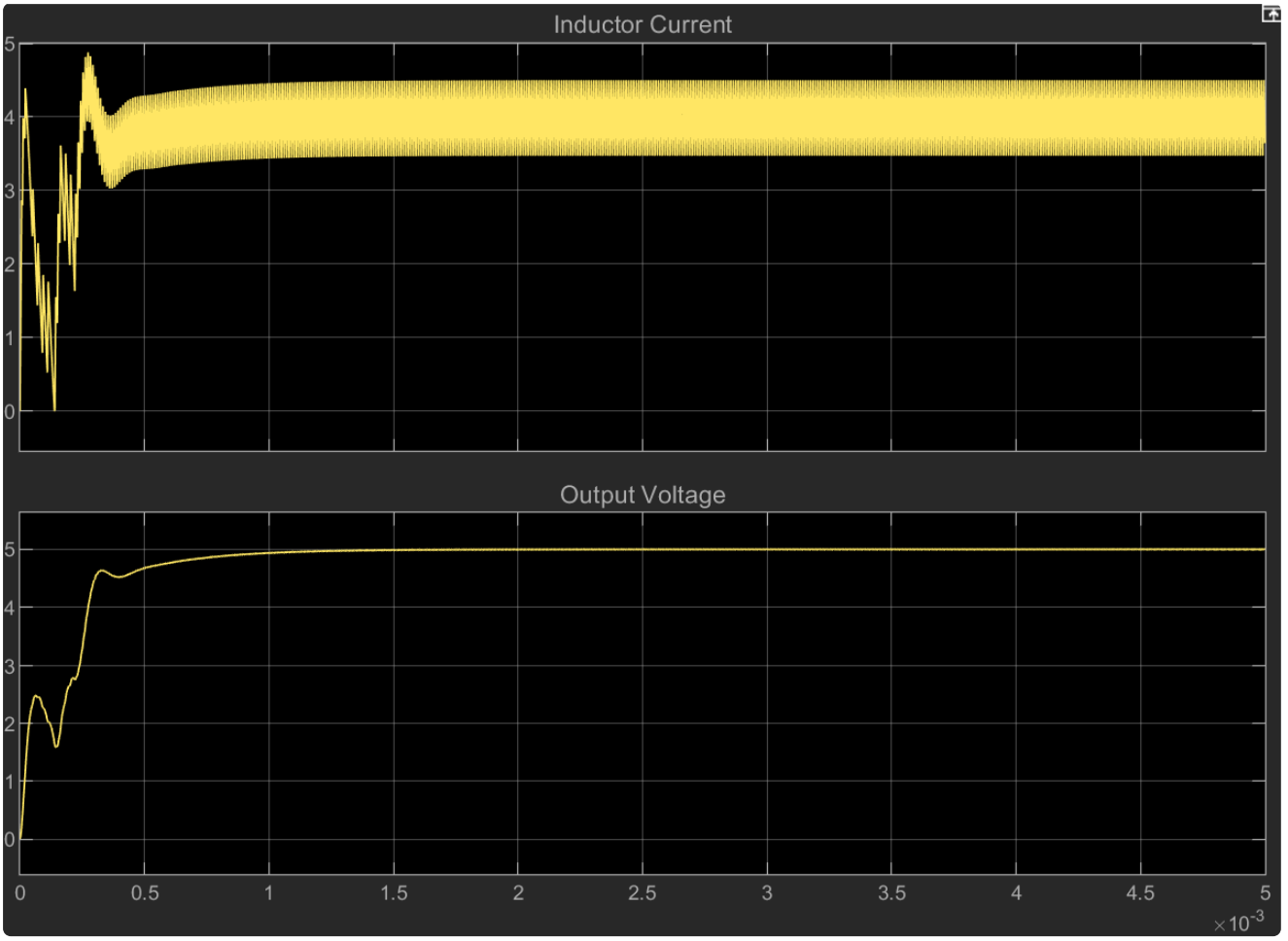


Figure 8. Inductor Current (A) and Output Voltage (V) in the closed-loop control system with input voltage of 14V (high side of range) and nominal load.

We can see that the controller operates well within the full desired range of input voltage, with settling time of about 1 ms and no overshoot for all of the cases checked.

6. Limitations

There are three main limitations of this design:

- Light load point is not so good, as it includes only ESR as damping.
- Controller is continuous time, whereas real implementation would add sampling and delay, which will result in decreasing the phase margin
- And finally, DCM is not covered. Below about 2.5W of output power, the converter enters DCM where the averaged CCM model is no longer applicable.

We can run some testing in the DCM region to observe the operation:

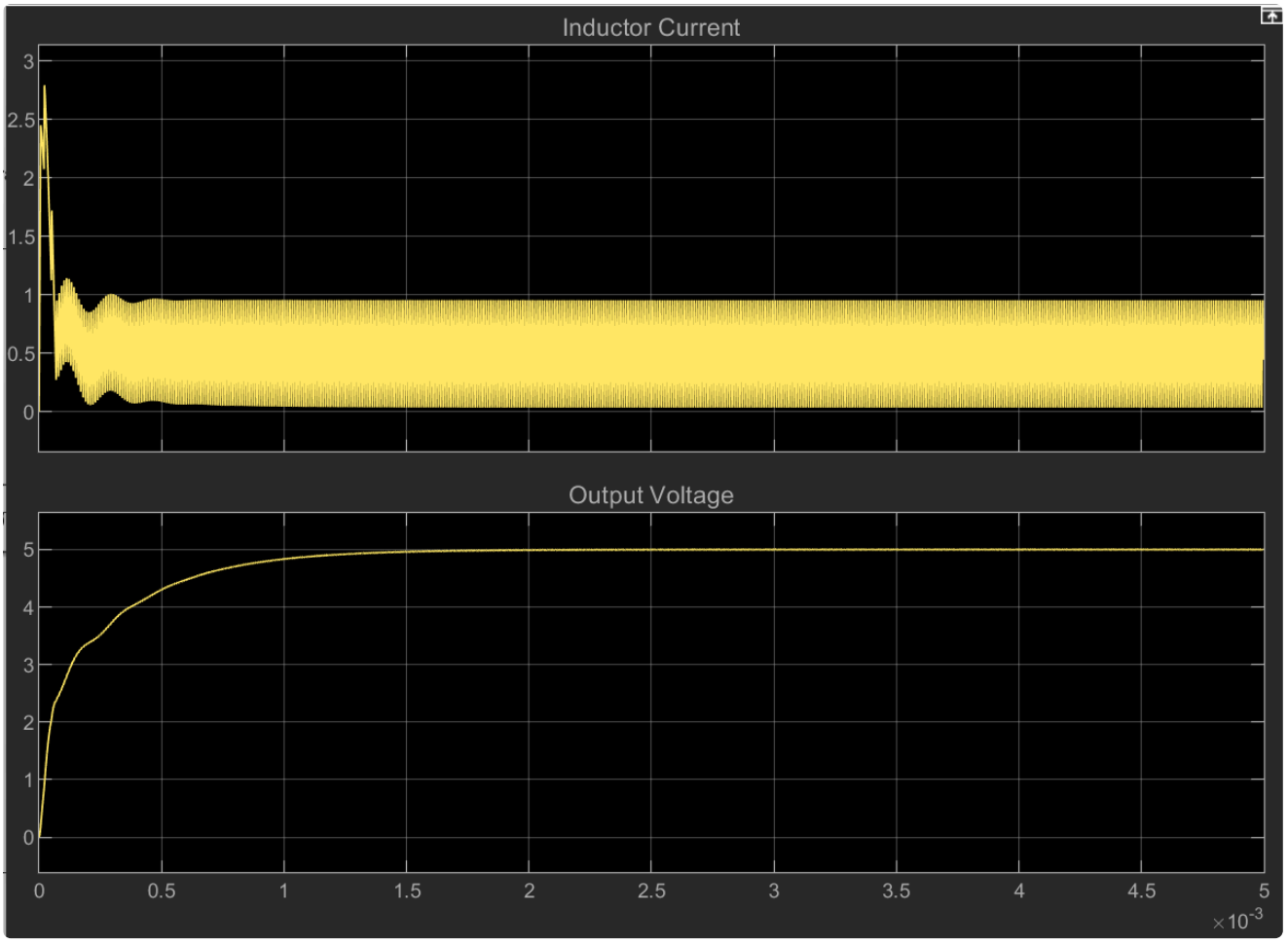


Figure 9. Inductor Current (A) and Output Voltage (V) in the closed-loop control system with nominal input voltage and output power of 2.5W (CCM boundary).

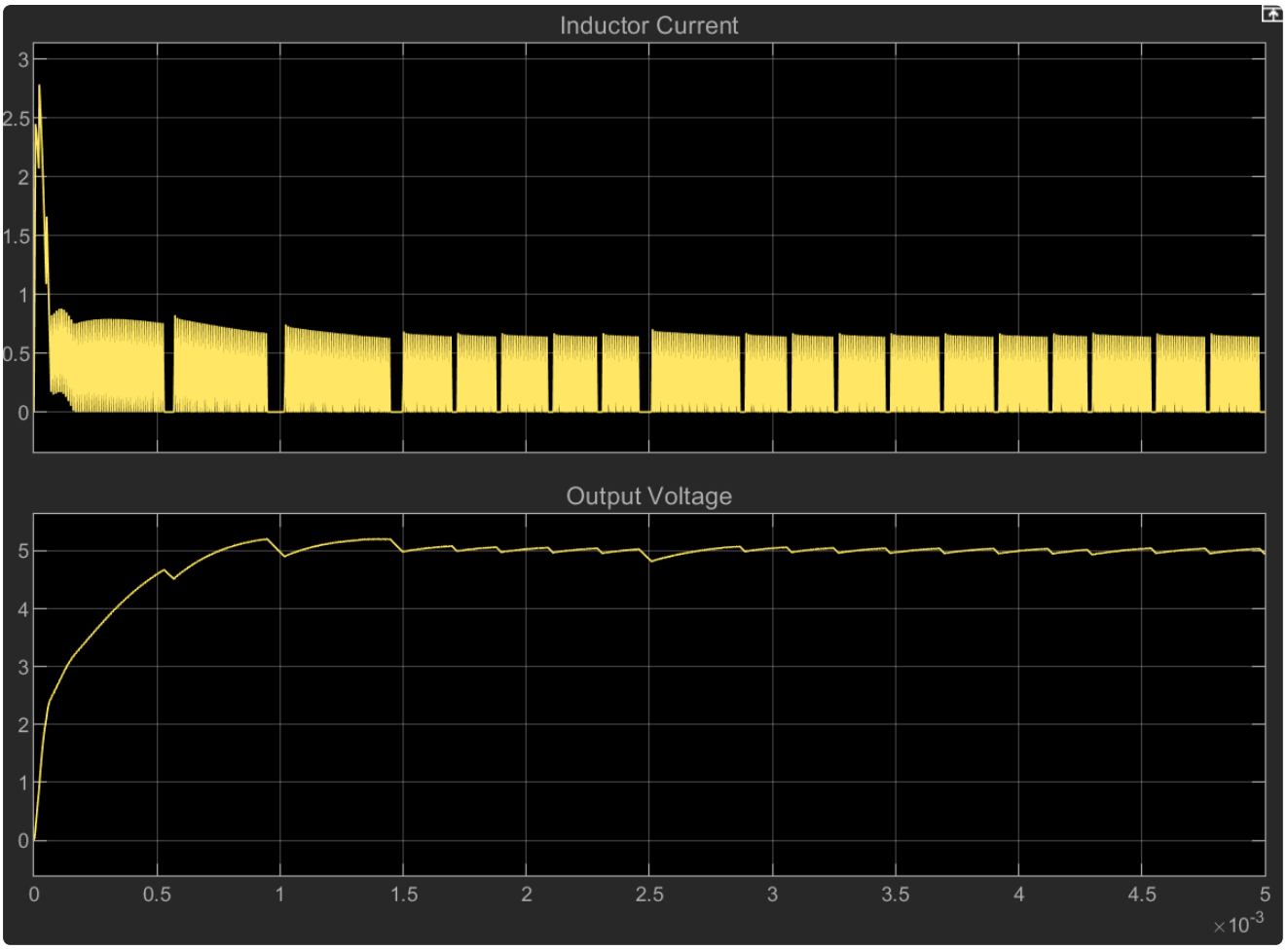


Figure 10. Inductor Current (A) and Output Voltage (V) in the closed-loop control system with nominal input voltage and output power of 1W (DCM).

We can clearly see the problems arising, and the output only gets worse with lighter loads.

7. Conclusion

A Buck Converter with a voltage-mode PID controller was designed from an averaged small-signal model and verified on a switching Simulink Simscape model. The load-dependent damping of the LC filter caused automatic tuning to fail, whereas manually placing a real double zero below the resonance made the control robustly stable across 10-14V input and 0.5-4A load.
The final controller operates well within the CCM and the designed range, with 1 ms settling time and no overshoot, though failing at lighter loads in DCM, as duty cycle and output voltage relation changes.

Possible improvements:

- Improve the controller to operate better in DCM, either by retuning the PID (easy but worse result), or more interesting approaches like gain-scheduling / current-mode control.
- Replacing the freewheeling diode with a MOSFET to reduce losses (synchronous buck).

8. References

Design of Power Management Integrated Circuits, Bernhard Wicht, 2024 - all the theory and used formulas can be found there, especially Chapter 10, which is devoted to Inductive DC-DC Converters.